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Nuclear Engineering – Politecnico di Milano

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ABSTRACT

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NOMENCLATURE

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CONTENTS

1 INTRODUCTION

1.1 Context

Italy is facing a significant energy transition, with the aim of achieving net-zero emissions by 2050. The residential sector plays a crucial role in this transition, as it accounts for $\sim 22\%$ of the total energy consumption and emissions [1].

The residential sector shows a high potential for decarbonization through the adoption of renewable energy sources such as solar photovoltaic (PV) systems and heat pumps (HP). However, this transition also introduces significant challenges in terms of monitoring and management, as it leads to a highly distributed production network that complicates national grid operations. Areas with dispersed populations and a high proportion of residential buildings often face particular difficulties during this transition. Solar photovoltaic systems are the primary choice for domestic renewable energy, but single-family houses are likely to feed a significant amount of their production into the grid, unlike apartment blocks of the same area where self-consumption is much higher.

In this context the role of nuclear energy is being reconsidered, especially with the advent of Small Modular Reactors (SMRs). Specifically, the present study focuses on the north-western part of the province of Varese, in the Lombardy region of Italy, where an hybrid system based on the integration of SMRs in a solar powered grid is being evaluated.

The integration of SMRs could provide a stable and reliable energy source to complement the intermittent nature of renewable energy sources such as solar PV systems, while also addressing the challenges of grid management and energy distribution in these areas.

1.2 Case Study

This study evaluates the technical and economic feasibility of integrating small modular reactors (SMRs) into the residential sector, considering the area's specific characteristics and the potential benefits and challenges of such implementation.

The proposed system consists of two SMRs, each with a nominal power output of 180 MWe, coupled with a secondary system capable of producing hydrogen through high-temperature steam electrolysis (HTSE). This system also includes synthetic gas production via CO2 methanation, which feeds into the gas mains and powers a small gas turbine with a nominal electrical output of 47 MWe. The electricity generated by this integrated system along with the residential PV production is designed to meet the entire demand of the service area. Additionally, the plant is configured to handle demand peaks and enhance profitability and adaptability to various energy market requirements.

1.2.1 Reactor

Our plant consists of two SMRs installed in Ispra, where the Joint Research Center (JRC) is located. The JRC is a research facility of the European Commission which has been chosen since it represents an already nuclearized site. This suggests the possibility of an easier pathway towards the construction of a future nuclear plant.

Moreover, the facility is home to the ESSOR reactor, currently facing decommissioning. The installation of at least one of the two SMRs could take place, technically, within the ESSOR reactor building. The containment was oversized due to the particular use case of the ESSOR reactor:

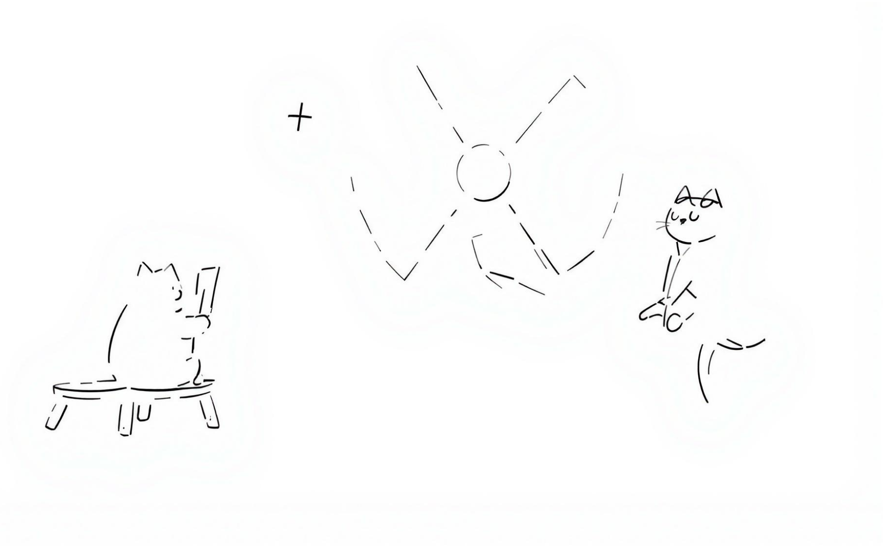


Figure 1: Simple schematic of the proposed hybrid plant integrating SMRs, hydrogen production, methanation, and residential PV.

primary and secondary circuits, as well as several laboratories are accommodated within the main building.

1.2.2 Secondary Plant

The secondary plant is designed for dual-purpose hydrogen utilization. Its primary function is to produce hydrogen through high-temperature steam electrolysis (HTSE), with a constant output allocated for market supply. Additionally, the plant uses the remaining hydrogen to produce synthetic gas via CO2 methanation. This synthetic gas production serves two main purposes: it fuels the gas turbine and creates a surplus that can be stored to reduce the region's dependence on fossil fuels, thereby supporting decarbonization efforts.

1.3 Objectives

This study aims to evaluate both the technical and economic feasibility of the proposed installation.

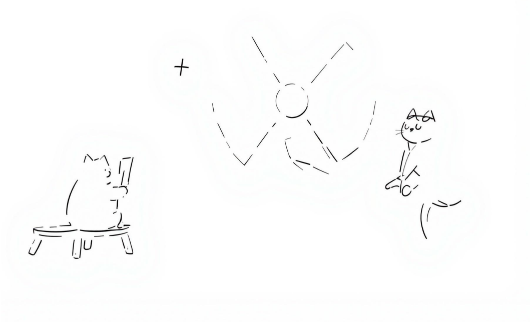
From a technical perspective, the system's capacity to meet energy demands and adapt to transient conditions will be assessed. From an economic standpoint, the total plant costs will be analyzed to determine the conditions required for achieving a positive net present value (NPV) within a reasonable timeframe.

The evaluation will consider four distinct scenarios:

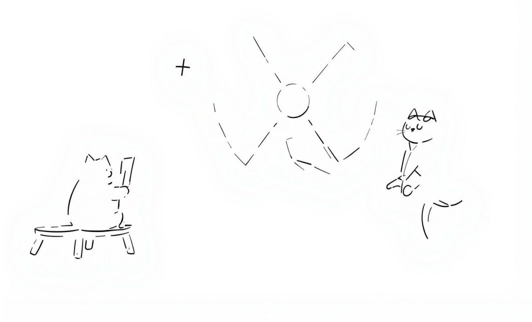
- Baseline Scenario: Current state of the residential sector in the area
- Optimistic Scenario: Residential sector is mostly electrified with highly efficient homes

- Pessimistic Scenario: There is still a significant share of fossil fuel heating systems in the residential sector
- Middle Ground Scenario: A middle ground between the optimistic and pessimistic scenarios, with a balanced mix of electrification and fossil fuel systems

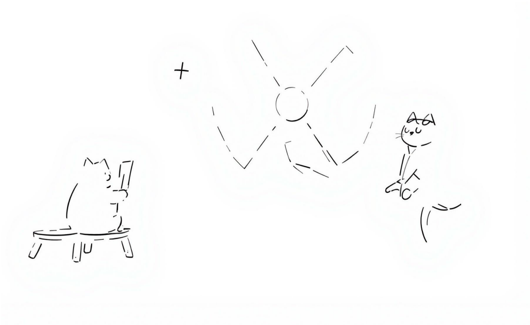
These scenarios will be represented as different distributions of energy classes within the residential sector.



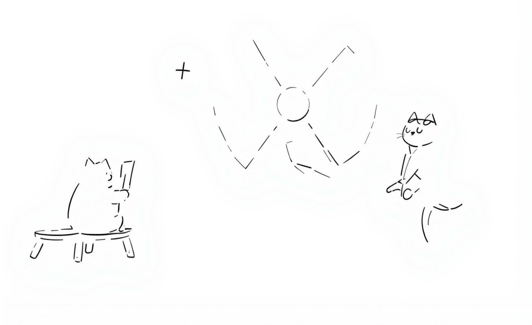
(a) Baseline Scenario



(b) Optimistic Scenario



(c) Pessimistic Scenario



(d) Middle Ground Scenario

Figure 2: Distribution of energy classes in the residential sector for the four scenarios.

The effectiveness of the hybrid system is evaluated with respect to the following key aspects:

- The investment cost to make the plant profitable
- The decarbonization impact
- The adequacy of the system's response to sharp transient conditions

1.3.1 Decarbonization Impact

The decarbonization impact will be assessed through an analysis of CO_2 emission reductions achieved by implementing the hybrid system. A comparative approach will be adopted, examining emissions from both the current residential sector energy mix and those projected for the proposed system across various scenarios.

The reduction will be quantified based on two primary factors:

- The volume of conventional gas replaced by CO_2-neutral synthetic gas
- Changes in the overall energy mix composition

Both aspects will be evaluated using grams of CO_2-equivalent per kilowatt-hour ($\frac{gCO_2-eq}{kWh}$) as the measurement unit. The decarbonization impact will ultimately be expressed as a percentage variation from the current state of the energy mix.

2 DATA COLLECTION

At the basis of this work there is the need to have a realistic energy demand profile for the residential sector in the region under study. Given our focus on the residential sector, the overall energy demand can be seen as the sum of the heating/cooling contribution + all the other electricity consuptions for appliances and lightning. Several semplifications will be made through the following derivation but we point out any possible improvment that could be made but was considered out of the scope of this work.

2.1 Approach

2.1.1 Introduction

In Italy, an APE (Attestato di Prestazione Energetica) is a document that certifies the energy performance of a propriety unit (*unità immobiliare*). This certification must be issued by a qualified technician and is mandatory for all buildings that are sold, rented or refurbished.

The APE classifies buildings in energy class, a letter grade (from A4 to G), that indicates the energy performance of the building, with A4 being the most efficient and G being the least efficient. The certificate includes many informations about the building among which it is possible to find information about geometry, plants, materials as well as the use of the building (commercial, residential, industrial, public, etc.).

Each unit refers to a distinct and identifiable part of a real estate property, such as an apartment, a house, or a commercial space, that can be individually owned, sold, or rented. For instance, a building with three apartments will have three separate APEs, one for each apartment. For this reason we can safely assume that each unit is the residence of a single family.

2.1.2 Methodology

The plan is to determine the demand profile of electricity in the region $P_{e,\text{tot}}$ from a distribution of energy classes. To do so it is necessary to first determine the reference demand profile for each energy class $P_{e,\text{ref},i}$, where i is the energy class.

For each energy class there is need to determine:

- Heating thermal demand $\dot{Q}_h$
- Cooling thermal demand $\dot{Q}_c$

- Domestic hot water thermal demand $\dot{Q}_{dhw}$

For each energy class we also need the share of heat pumps $\chi_{HP,i}$ to differentiate between electrified and fossil based heat generation. The primary energy demand (electricity or fossil fuels) are obtained in a simplified manner with average efficiencies.

Regarding the electricity production from photovoltaic system we need to determine the share of buildings with photovoltaic systems for each class $\chi_{PV,i}$ and the average size of the photovoltaic system $P^{+}_{e,i}$ for each energy class, this last was differentiated between households with and without an heat pump. A reference profile of the PV production for the region is obtained and then weighted on the simulated scenario.

Another electricity contribution is the one from appliances and lighting, $P_{e,other}$ which is assumed to be equal for all classes.

$$
\begin{aligned}
P_{e,\text{ref}} = \sum_{i=\text{A4}}^{\text{G}} \Bigg[& \chi_{\text{HP},i} \cdot \left(\frac{\dot{Q}_{h,i}}{\text{COP}} + \chi_{\text{cooling}} \cdot \frac{\dot{Q}_{c,i}}{\text{EER}} \right) \\
& - \chi_{\text{PV},i} \cdot \left(\chi_{\text{HP},i} \cdot P_{\text{PV,w/HP}} + (1 - \chi_{\text{HP},i}) \cdot P_{\text{PV,w/oHP}} \right) \\
& + P_{e,\text{other}} \Bigg]
\end{aligned} \tag{1}
$$

Then, this can be multiplied by the number of buildings in the region N_{UI} to obtain the total demand $P_{e,\text{tot}}$.

$$
P_{e,\text{tot}} = N_{\text{UI}} \cdot P_{e,\text{ref}} \tag{2}
$$

2.1.3 Data Sources

From the CENED database [2] we extracted the information about the relevant municipalities of the region, which we derived based on the map of primary substations from the national grid operator (GSE) [3]. The data was also filtered to include only residential buildings, excluding commercial and industrial ones. Furthermore, we filtered unhealthy data points at every step of the process, such as buildings with areas larger than $1000m^2$ (mostly releated to errors in unit of measurement or lack of care in the production of the APE). This was necessary given the large amount of data from the complete regional database. This way an easier to manage csv file was obtained. The number of buildings was computed from the ISTAT database [4] by extracting the population of the municipities and dividing by the average household size in the province.

2.2 Heating and Cooling Demand

For each energy class we have taken a real example of a residential building from the region under study and computed the hourly thermal demand for heating and cooling thoughout the year. This is obtained by means of a professional software (TERMOLOG) that allows for dynamic simulation of the energy system of the building. This computation is based on the UNI EN ISO 52016 standard. We used one reference building per energy class all with the same utlization profile. The output of the calculation is an hourly profile of the thermal demand for heating and cooling.

2.3 Domestic Hot Water Demand

The domestic hot water demand is assumed to be equal for all energy classes. It has been computed based on the consuption profile depicted in [5], as a reference we choose the profile of swizerland since it is similar to the one of the region under study. It is then scaled based on the average number of people per household in the region, which is 2.25 [4]. This thermal demand is then summed to the heating demand profile.

2.3.1 Normalization of the Data

Given that our data is based on real building we had to normalize the data to obtain a reference value for each energy class. This is done by taking into account the userful heated area of the building and the EP,nren (Primary Energy Demand for non-renewable energy) value of the APE. The EP,nren is the value used to determine the energy class of the building and is defined as energy per unit of area per year. Therefore we multiply the EP,nren by the useful heated area of the building to obtain the total primary energy demand for non-renewable energy. Convertion factor for each energy class are taken as the ratio between said value obtained by the averages extracted from the CENED database [2] and that for our reference buildings.

$$C = \frac{\text{EP,nren}_{\text{CENED}} \cdot A_{\text{r,CENED}}}{\text{EP,nren}_{\text{REF}} \cdot A_{\text{r,REF}}} \tag{3}$$

2.4 Heat Pump distribution and Cooling Demand

We identified the share of heat pumps per energy class. We also determined the share of buildings that have a cooling system installed, which as of today comes up to 9.58%. This is a very low share and is to be expected as the region under investigation is quite chill during summer.

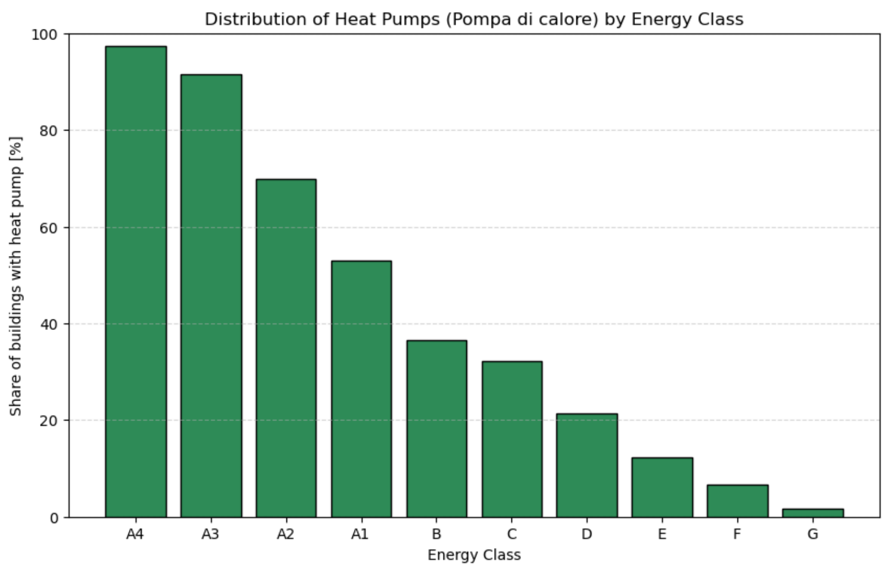


Figure 3: Distribution of heat pumps across energy classes.

2.5 Photovoltaic

2.5.1 Photovoltaic Distribution

We identified the share of building that have photovoltaic systems installed per energy class.

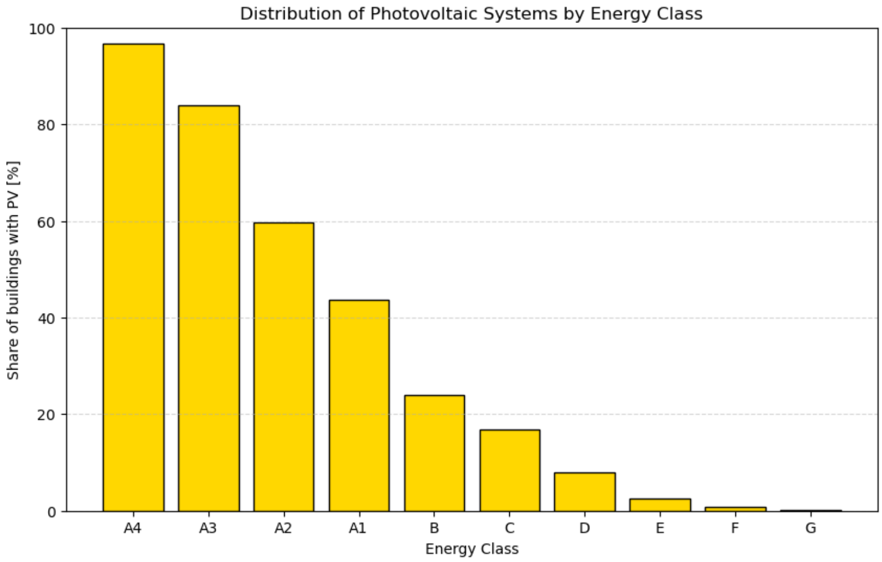
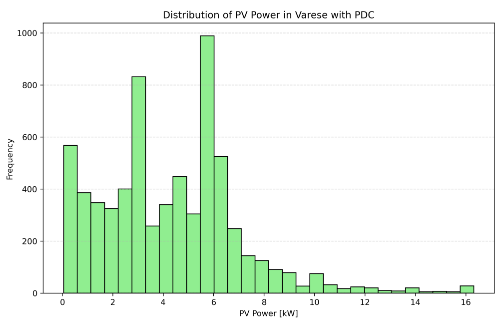


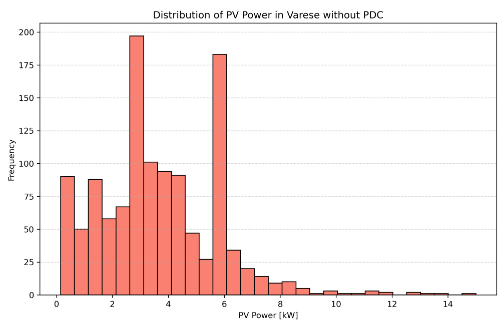
Figure 4: Distribution of photovoltaic systems across energy classes.

2.5.2 Photovoltaic System Size

Moreover we then divided the data into two groups, those with an heat pump and those without. From these we plotted the distribution of the size of the photovoltaic system installed. And computed the average (of the P_99)



(a) Photovoltaic system size distribution for buildings with heat pumps.



(b) Photovoltaic system size distribution for buildings without heat pumps.

Figure 5: Distribution of photovoltaic system sizes based on heat pump presence.

The averages values are: 4.30kW and 3.70kW for the sistem with and without heat pumps respectevely.

2.5.3 Reference Production

We have to obtain a reference value of the photovoltaic production profile. First, the list of municipalities is fed to the Nominatim API [6] to obtain the coordinates of the center of each municipality. Then, we use the PVGIS API [7] to obtain the hourly production profile for each municipality. We fixed the angle to an average $22°$ and a plant size of $1kW$. For each municipaly we obtain a profile every $10°$ of the azithmut angle, from $0°$ to $360°$. Those are then weighted to a gaussian distribution to represent that it is prefered to point the photovoltaic system to the south. This gives us a reference photovoltaic production profile for each municipality $P^{+}_{e,m}$, where m is the municipality. Then we can obtain the reference production profile for each energy class $P^{+}_{e,ref}$ by weighting the production profile of each municipality on the population in that municipality [4].

2.6 Other Electricity Demand

Electrical demand for appliances and lighting is modeled equal for all classes. The definition was manual, hour by hour for an average day in the following ”seasons”:

- Winter Weekday
- Winter Weekend
- Summer Weekday
- Summer Weekend
- Winter Holiday
- Summer Holiday

Moreover we added an absence factor to take into account when people go on holiday massively during august.

The profile is defined by considering the following contributions:

- standby and always on appliances
- lighting
- appliances

During the summer the need for a cooling fan was added. Moreover we added a parameter to rapresent the share of buildings with an induction system installed.

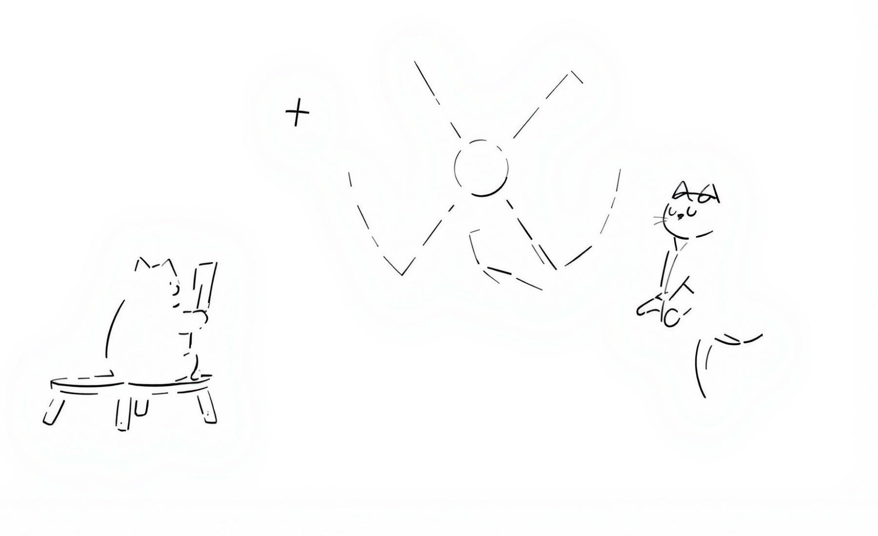


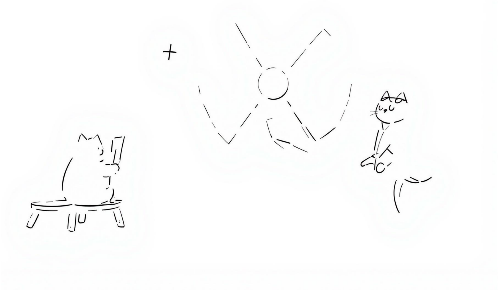
Figure 6: Reference hourly profile through the year.

2.7 Validation

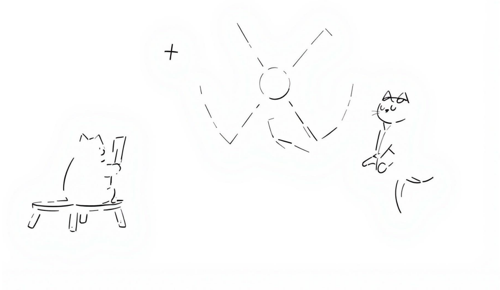
To verify our method is acceptable we compare the total demand by the data available online, in particual we compared it to the average annual deamnd, in the residential sector for the province of Varese [8], which happends to be $1931kWh$ in 2023, with a decreasing trend in the last years ($2016kWh$ in 2022 and $2119kWh$ in 2021).

2.8 Demand Distribution

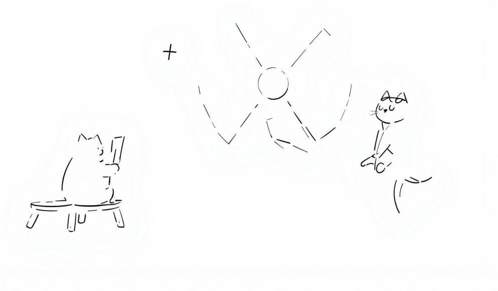
To better visualize the behaviout of the demand we plot the distribution of the demand hour-by-hour for each scenario.



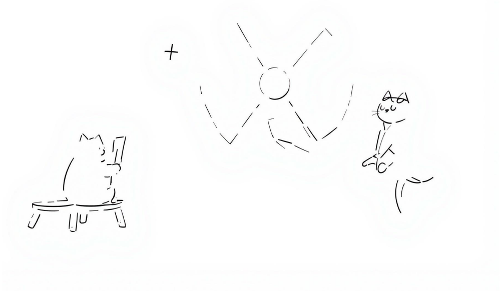
(a) Scenario 1



(b) Scenario 2



(c) Scenario 3



(d) Scenario 4

Figure 7: Distribution of the demand hour-by-hour for four scenarios.

3 TECHNICAL PLANT MODELLING

The whole plant was modeled in Dymola, a technical software using the Modelica language for the dynamic analysis of complex systems. The model is based on the hybrid plant concept, which combines two 177 MW_e Small Modular Reactors (SMR) with a hydrogen production plant and a methanation reactor producing CO_2-neutral syngas for a gas turbine. The TANDEM project library (reference) was widely used to model the enritre plant, with the exception of the methanation reactor, which was modeled from scratch in Dymola.

3.1 Main assumptions and plant layout

To meet the overall energy demand of the target region, the plant design combines two 177 MW_e Small Modular Reactors (SMRs) with fifteen High Temperature Steam Electrolyzers (HTSEs), a Methanation Reactor and a small Gas Turbine (47 MW_e). However, to reduce the computational burden of the simulation in Dymola, the system is modeled using a single SMR and a single HTSE module. The actual electricity output from the SMR and hydrogen production from the HTSE are then scaled by multiplying the simulated values by the respective number of units within the model. These scaled values are used to evaluate the overall energy balance and to feed the syngas production module.

The SMR supplies electricity to the grid, as well as both electricity and the required heat to the hydrogen production unit. The HTSE generates hydrogen from water using the thermal and

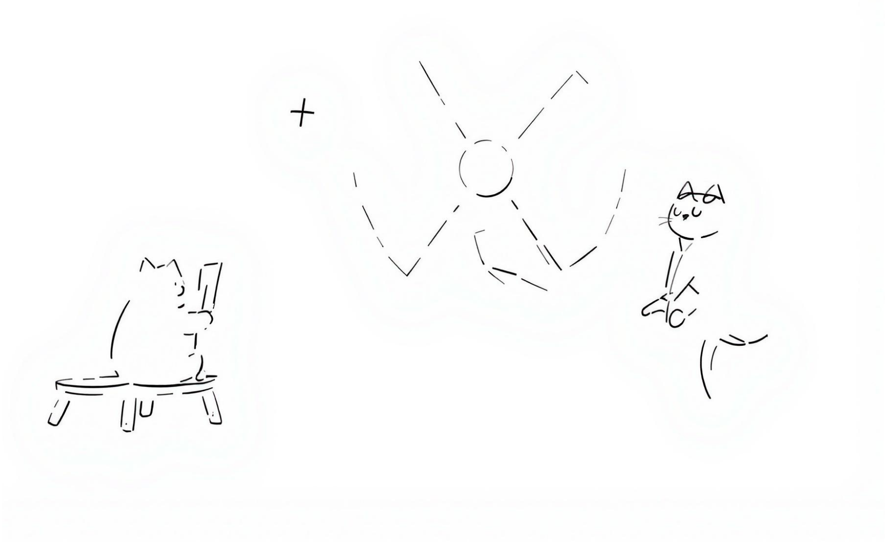


Figure 8: Simple schematic of the proposed hybrid plant in Dymola.

electrical energy provided by the SMR. At each time step, 10% of the produced hydrogen is assumed to be stored temporarily in a buffer for future sale. The remaining 90% is sent to a methanation reactor, where it reacts with carbon dioxide to produce methane.

The resulting syngas is used as fuel in the gas turbine, while any surplus is stored in a buffer tank. This stored syngas can later be used to cover the residual fossil fuel demand of the region (i.e., the remaining heat demand not met by other sources).

3.1.1 SMR

The TANDEM library [9] was used to model the SMR, employing a simplified representation of a Small Modular Reactor. Despite being a simplified model, the overall system is quite detailed and includes various components, allowing for a realistic simulation of the reactor's behavior. It captures the dynamics of the primary and secondary loops, the turbine, the condenser, and the associated control systems.

The selected model is a 177 MW_e Light Water Reactor (LWR), originally designed for cogeneration, providing both electricity and thermal energy. To support this dual purpose, the reactor features three steam extraction points — High Pressure (HP), Intermediate Pressure (IP), and Low Pressure (LP) — which allow heat to be drawn from different stages of the turbine. In the present configuration, the heat required for the HTSE is extracted via the IP tap, immediately downstream of the HP turbine stages. The HP and LP extraction lines are disabled and manually set to zero, as they are not utilized in this setup.

Some useful specifics of the SMR model are:

- Thermal power: ... MW_{th}
- Electrical power: 177 MW_e
- Thermodynamic Efficiency: ...

- ....
- ....

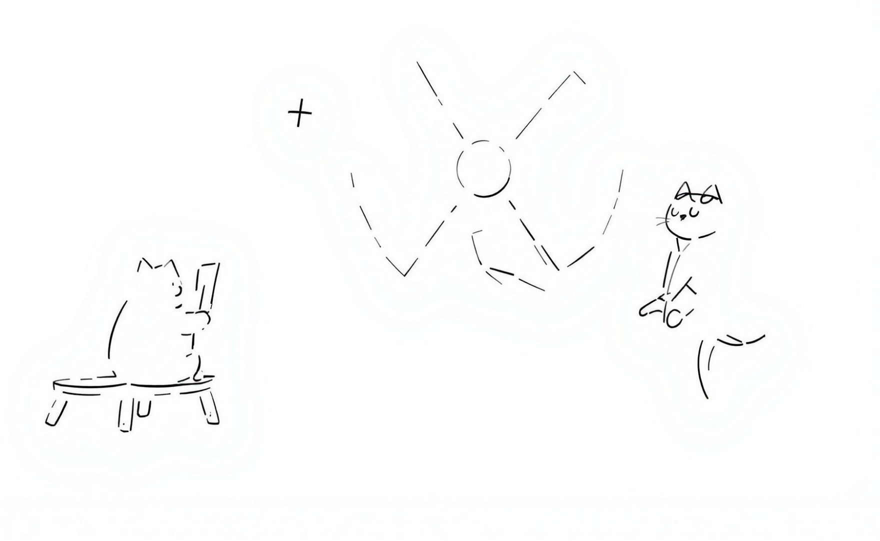


Figure 9: Simple schematic of the selected SMR model in Dymola.

3.1.2 H2 Plant

The hydrogen production plant is modeled using the **TANDEM** library [9], which provides a detailed representation of a High Temperature Steam Electrolyzer (HTSE). The HTSE is designed to produce hydrogen from water by utilizing both thermal and electrical energy supplied by the SMR. The model captures the dynamic behavior of the HTSE, including its thermal and electrical inputs, and simulates the hydrogen production process accordingly.

The model receives as input a target hydrogen production rate—defined in the demand section of the system—and the thermal energy extracted from the SMR's steam tapping point. A key modeling assumption is that the heat input to the HTSE corresponds to the `Real` value of the heat exchanged in the Heat Exchanger connected to the Intermediate Pressure (IP) tap loop of the SMR, and thermally linked to the HTSE model.

Based on this thermal input, along with several operational parameters, the HTSE model calculates the corresponding hydrogen production, resulting in a computed mass flow rate of H_2. An estimate of the energy consumption associated with hydrogen production is also computed and subsequently included in the overall energy balance of the system.

Some key characteristics of the HTSE include:

- operating temperature?
- operating pressure?
- efficiency?

- electricity consumption?
- production rate?

The H_2 plant is equipped with a buffer tank used to temporarily store the portion of hydrogen intended for sale, while the remainder is sent directly to the methanation reactor.

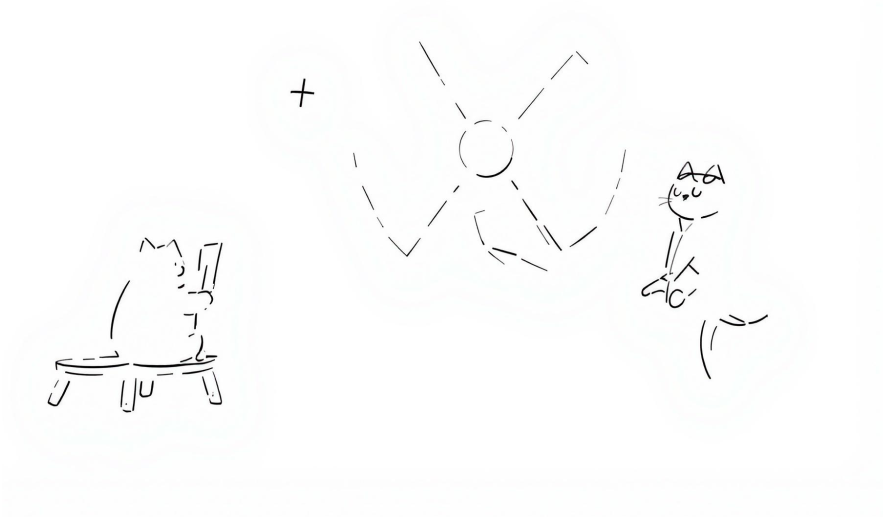


Figure 10: Simple schematic of the H_2 plant model in Dymola.

3.1.3 Methanation Reactor Modeling Approach

The methanation reactor was modeled to simulate the conversion of hydrogen (H_2) and carbon dioxide (CO_2) into methane (CH_4) and water (H_2O) through the Sabatier reaction:

$$\mathrm{CO_2} + 4\mathrm{H_2} \rightarrow \mathrm{CH_4} + 2\mathrm{H_2O} \qquad \Delta H = -165.1\ \mathrm{kJ/mol} \tag{4}$$

To estimate the amount of CH_4 produced from a given reactant feed, a dimensionless parameter, $conversionEfficiency$, was introduced. This parameter (default value: 0.78) represents the maximum chemical efficiency of catalytic methanation, as reported by Götz et al. [10]. It defines the fraction of the limiting reactant (either CO_2 or H_2) that is effectively converted into CH_4, accounting for real-world inefficiencies and the incomplete conversion typical under practical temperature and pressure conditions.

The molar output flows are then determined based on reaction stoichiometry:

$$\dot{n}\mathrm{CH4} = \dot{n}\mathrm{lim} \cdot \eta \; \dot{n}\mathrm{H2}O \qquad = 2 \cdot \dot{n}\mathrm{CH4} \; \dot{n}\mathrm{H2} = \dot{n}\mathrm{H2,in} - 4 \cdot \dot{n}\mathrm{CH4} \; \dot{n}\mathrm{CO2} \qquad = \dot{n}\mathrm{CO2,in} - \dot{n}\mathrm{CH_4}$$

The model is reported in the appendix, along with the other components developed from scratch in Dymola.

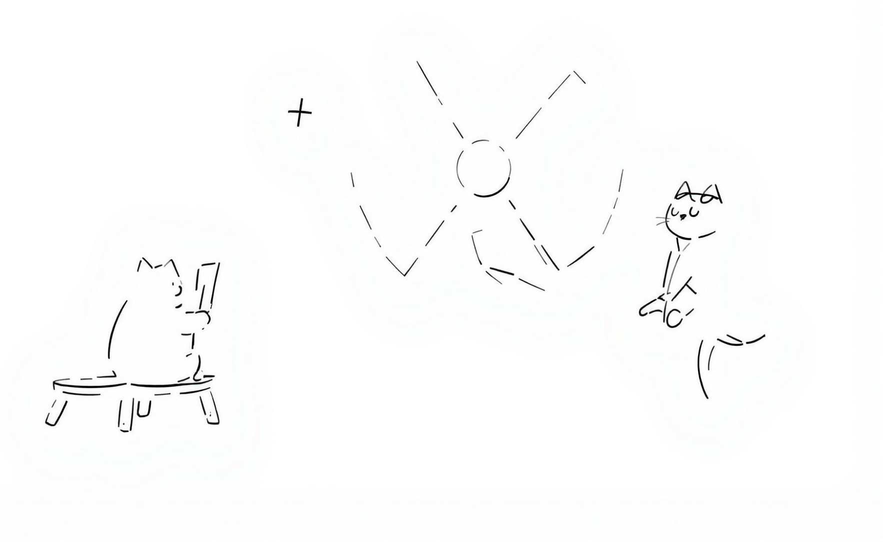


Figure 11: Simple schematic of the Methanation Reactor model in Dymola.

3.1.4 Gas Turbine

3.1.5 H_2 and CH_4 buffers

3.2 Control Room

3.3 Demand Definition

To run the model we decided to compute a priori the daily setpoint for:

- Turbine load, limited between 35% and 100% of the maximum load
- HTSE load, limited between 60% and 100% of the maximum load
- Steam tap flow from the intermediate pressure of the SMR.

We first of all evaluated the electricity available for the H2 plant, from which we determined the load of that the H2 plant can take.

$$P_{e,available} = P_{e,SMR} + P_{turbogas}(35\%) - P_{e,grid} load_{H2} = \left(\frac{P_{e,available}}{214} + 0.197775\right) \cdot \frac{1}{N_{HTSE} \cdot 0.1} \quad (5)$$

Where the relationship $\dot{m_{H2}} = \frac{P_{e,available}}{214} + 0.197775$ was found by running a sweep simulation of the HTSE model in Dymola. The mass flow rate of hydrogen produced $\dot{m_{H2}}$ is the divided by the nominal mass flow rate produced by one module $0.1kg/s$ multiplied by the number of modules N_{HTSE}. This load had then to be clipped for values between 60% and 100%, where 60% is the lower operational limit.

Thereafter we estimated the syngas production rate which from the first simulation appeared to be

3.3.1 Dati che diamo in input

a

3.3.2 Rate limiting

a

3.3.3 Clusters

spiega che famo i clusters + cita che usiamo dei profili di alcuni gioni particolari per vedere rispospta del sistema

4 ECONOMIC STUDY

5 RESULTS

5.1 Scenario 1

distribuzione enrgy classes profilo DOMANDA bilanci finali usciti da dymola balance economico risposta ai transients key psarameters

6 POSSIBLE IMPROVEMENTS

6.1 On the Data Collection

It is defenitly possible and could give better results if the data for each class was obtained by a set of real building from the region instead of just one. Ideally one would use a statistically signfinicant set of buldings and vary the utilization profile as well. Moreover, it would be beneficial to wheight the simulation data on the real energy requirments of those same buildings, which is a much more complex task since it would require extensive data acquisition during the year. While this is feasible for electrical demand (most operators give to the customer a detailed bill with hourly consumption), it is much more difficult to get the same data for fossil fuel consumption. At last, the normalization has been done with the heated area an not with the volume (which is much more indicative of the energy requirment of a building) given the on-field knowledge that the volume is usually approximatly obtained by a the rule of thumb of multiplying the heated area by a factor of 2.7 or 3 in most cases. From the simulation we have neglected humidification and dehumidification electricity needs since these plants are rearly installed today, a better evaluation would evaluate the possible future share of air treatment plants in the residential sector.

Photovoltaic: We have neglected any solar-thermal system installed since their are considered not very impactful and their impact is complex to evaluate.

Convertion thermal to primary: use time dependent efficiency for heat pumps that take into account the ambient temperature, this could be accomplished considering that our code uses hourly data.

Other Electricity Demand: This is without a doubt the most complex contribution to evaluate, expecially considering that it is not easy to imagine how the electricity consumption will evolve in 25 years. It could be improved by collecting a statistically signficant number of examples from the region on the type of appliances installed and their consumption.

Overall: the methodology of using class distribution as a way of evaluating the energy demand is considered a good approach, for the best results we shall gather consumption data for both electric and fossil fuels for a statistically significant number of buildings for each energy class.

6.2 On the Plant Modelling

6.3 On the Economic Study

7 CONCLUSION

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